

BOUZIR MOHAMED ALI

Data-Oriented Computer Engineer



Professional Summary

Computer Engineer specialized in Artificial Intelligence, Data Science, and backend development. Skilled in building scalable, data-driven systems that automate workflows, deliver insights, and improve decision-making.

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Portfolio: <https://dali-eight.vercel.app>

Education

École Supérieure Privée d'Ingénierie de Monastir

Engineering Program in Computer Science – Final Year (Specialization: Artificial Intelligence and Data Science)
2023 - Present

Faculty of Sciences of Monastir

Bachelor's Degree in Software Engineering and Computer Science (LGLSI)
2020 - 2023

Fattouma Bourguiba High School, Monastir

Baccalaureate - Technical Section (Honors: Good)
2019 - 2020

Professional Experience

ElyosDigital Company - Web Development Intern

Monastir, Tunisia | June 2024 - July 2024

- Established and launched a full-stack gym management platform in Laravel to digitize memberships, subscriptions, and coach scheduling, boosting administrative efficiency by approximately 30%.
- Introduced automated session-scheduling workflows that **reduced** manual coordination time by 40% and **increased** schedule accuracy by 25%.
- Streamlined coach assignment and session planning, increasing schedule accuracy and minimizing conflicts.
- Delivered responsive, secure back-office modules focused on data protection and user experience.

MyMatch - Backend & Admin Panel (Freelance Project)

Remote – Monastir, Tunisia | September 2024 - May 2025

- Designed and developed a centralized admin dashboard managing **7,000+ player profiles** and **70+ sports complexes**, improving data accessibility by **65%**.
- Built scalable Laravel backend services integrated with Firebase and REST APIs, achieving 99.9% uptime and reducing sync latency to under 2 seconds.
- Created **interactive dashboards** with real-time analytics, cutting manual tracking time by **40%**.
- Optimized **data flow and authentication logic**, enhancing system responsiveness and administrative productivity by **over 60%**.

Projects

AI Business Agent - Intelligent Decision-Support SaaS

- Architected a production-grade AI platform that integrates LLM reasoning, RAG retrieval, and Monte Carlo simulation for advanced business analysis.
Built a multi-route orchestration engine that handled over 24K queries and 150 simulations, achieving 97% routing accuracy with a 2.7-second median latency.
Connected PostgreSQL + pgVector with OpenSearch for hybrid semantic retrieval and deployed Grafana for end-to-end observability and performance metrics.
Tech: Next.js 14, FastAPI, PostgreSQL + pgVector, OpenSearch, MinIO, Redis, Llama 3.1 8B / GPT-4o-mini, Docker Compose, Grafana, Prometheus
Links: <https://www.kaggle.com/code/mohamedalibouzir/kaggle-llama31-refined/> <https://github.com/dalibouzir/AI-Agnet>

AFFA - Automated Fantasy Football Assistant

- Engineered an AI-based prediction engine using Random Forest and Bayesian inference, improving weekly player recommendation accuracy by 73% and reducing forecast error by 2 points per gameweek.
Analyzed 15+ gameweeks, achieving ±2-point forecast precision and reducing prediction error by 73% through MLflow-driven optimization.
Integrated real-time dashboards in Elasticsearch and Kibana for continuous model performance tracking.
Tech: Python, Flask, FastAPI, MongoDB, API-Football, Docker, MLflow, Elasticsearch, Kibana
Link: <https://bouzirdaliaa.pythonanywhere.com>

QuirkHire - AI Resume Recommendation Platform

- Created an LLM-powered talent-matching system leveraging NLP and hybrid recommender models.
Matched 1,200+ candidates across 200+ job posts, boosting recruiter match accuracy by 28% and cutting screening time by 35%.
Delivered a recruiter dashboard to monitor recommendation accuracy, model performance, and engagement analytics.
Designed a modern React interface improving user retention by 22% and overall application flow satisfaction.
Tech: React, Django, DRF, Supabase, Redux Toolkit, OpenRouter API
Link: <https://career-reco.vercel.app>

Meriem Booking - Web Platform for Therapists

- Developed a responsive booking platform using Next.js 14 and Supabase, enhancing therapist scheduling and client experience.
Implemented real-time calendar synchronization, session exception handling, and dynamic availability tracking.
Built an intuitive admin dashboard to manage appointments and therapist timetables efficiently.
Tech: Next.js 14, Supabase, motion-dom, Tailwind CSS
Link: <https://fittrahmoms.com>

Languages

- Arabic: Native
- French: Fluent (B2 - TCF Campus France)
- English: Fluent

Technical Skills

Programming Languages: Python, Java, C/C++, SQL, JavaScript, PHP, R
Web Development: React, Next.js, Laravel, Flask, Django, HTML, CSS, Bootstrap, REST APIs, Firebase, Supabase, Docker
Data Science & AI: Pandas, NumPy, Scikit-learn, TensorFlow, Keras, PyTorch, NLTK, spaCy, Transformers, VAE, Diffusion Models, CRISP-DM
Data Engineering & MLOps: FastAPI, MLflow, Elasticsearch, Kibana, PostgreSQL (pgvector), OpenSearch, Docker
Tools & Environments: Git, GitHub, VS Code, Postman, Airtable, Trello, Tableau, Excel, Oracle DB, Arduino IDE, Google Workspace

Soft Skills

Self-motivated, detail-oriented, and organized professional with strong analytical, problem-solving, adaptability, communication, time management, and collaboration skills in dynamic environments.

Certificates

- Python for Data Science, AI & Development — IBM (Dec, 2023)
- Fundamentals of Visualization with Tableau — University of California (Dec, 2023)
- What is Data Science? — IBM (Dec, 2023)

Interests

Music (piano), digital media, technology exploration, web research, and football.